Practical Digital System Design　Workstation Usage Process Worksheet

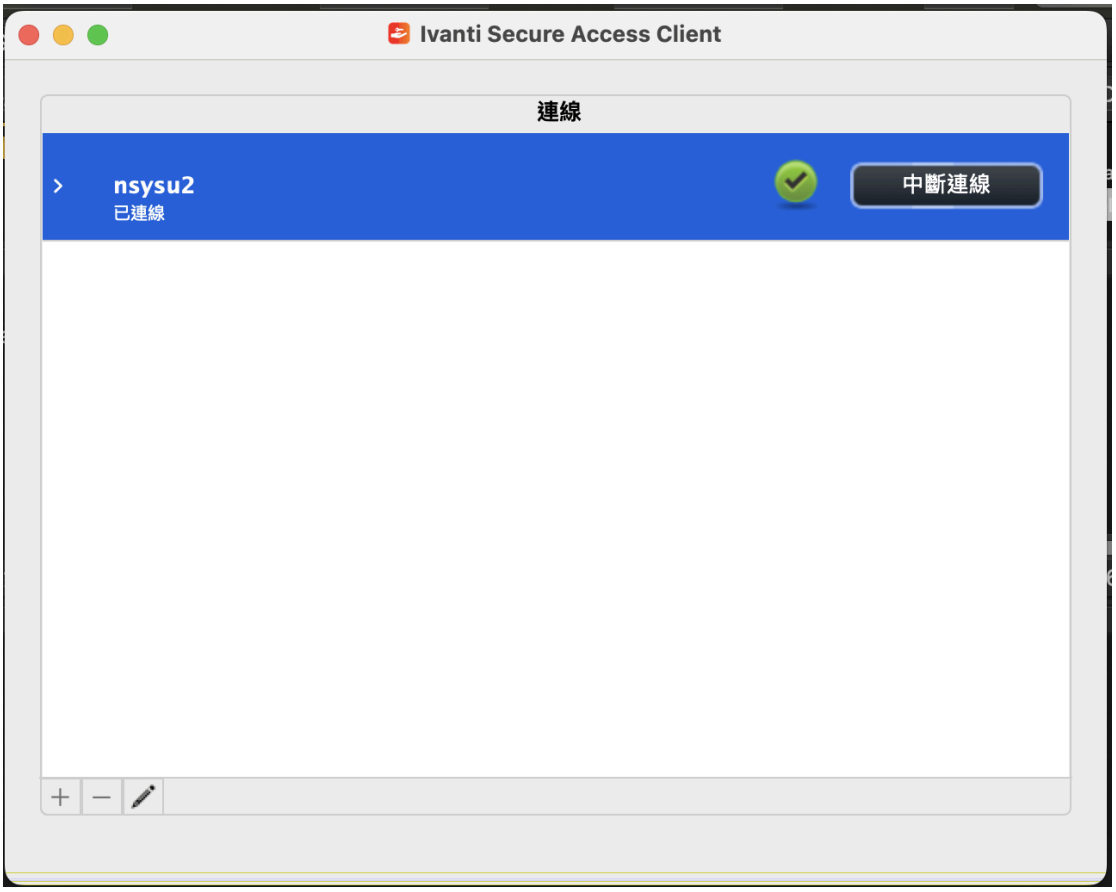
Student ID : B133015012

Name : EnPing Chung

Based on the workstation usage process introduction in handout. Please answer the following questions.

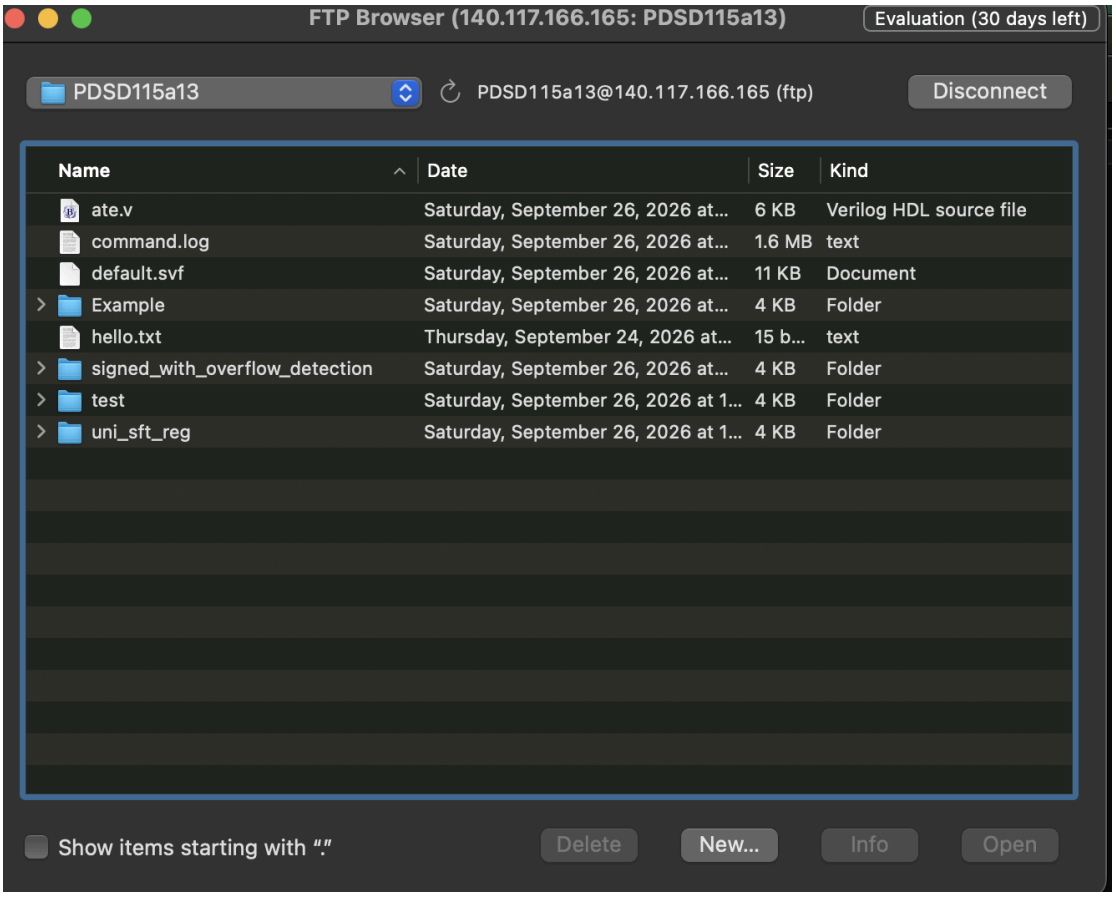
【Phase 1】: Setup and connection to the workstation

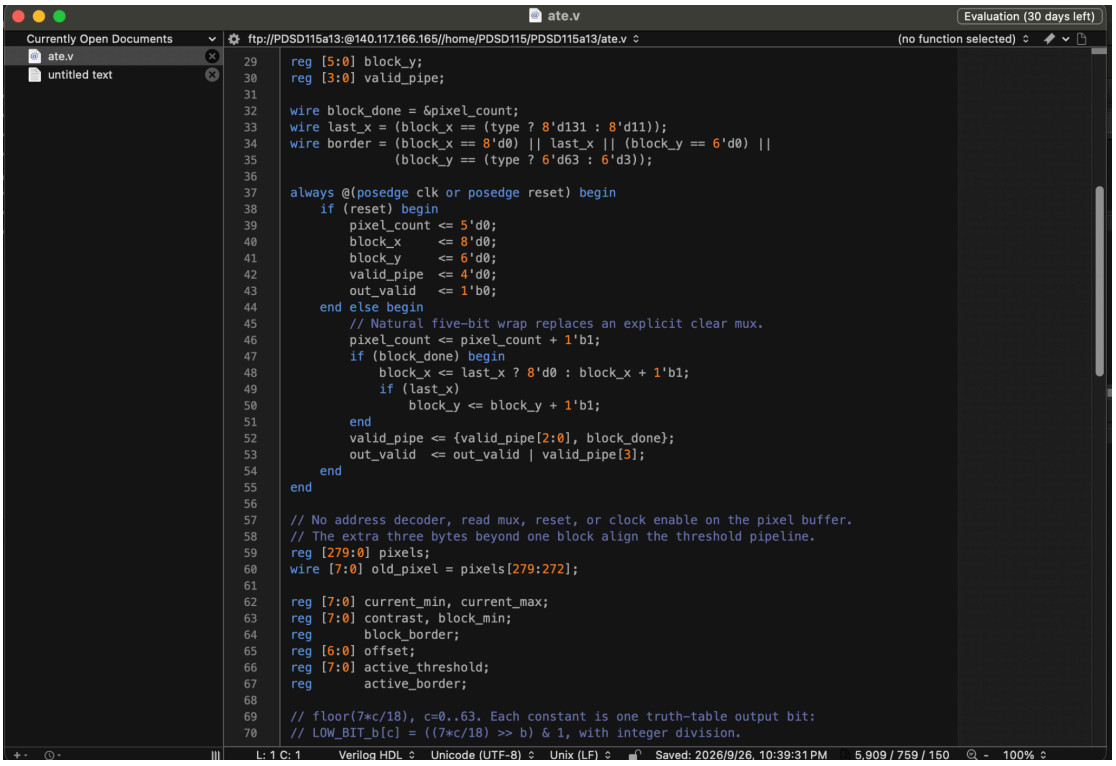
1. According to the VPN usage instructions from the Office of Library and Information Services, after completing the setup of Invanti Secure Access Client, can the program icon be seen in the taskbar on your local machine, and does it show as connected? Yes



(Also, please include a screenshot.)

2. According to pages 5 to 9 of the “03-I SOC Server Overview” handout, after completing the Notepad++ setup, can you see the folders and files of your workstation account in the NppFTP Window?
Npp is not available on MacOS so I am using BBedit





(Also, please include a screenshot.)

3. 3. According to pages 10 to 14 of the “03-I SOC Server Overview” handout, after completing the Xming setup, can the program icon be seen in the taskbar on your local machine?
Xming is not available on MacOS so i am using Xquartz





(Also, please include a screenshot.)

4. According to pages 15 to 22 of the SOC Overview handout, after completing the Xshell setup, can you see the TARS logo and your workstation account in the terminal? ____



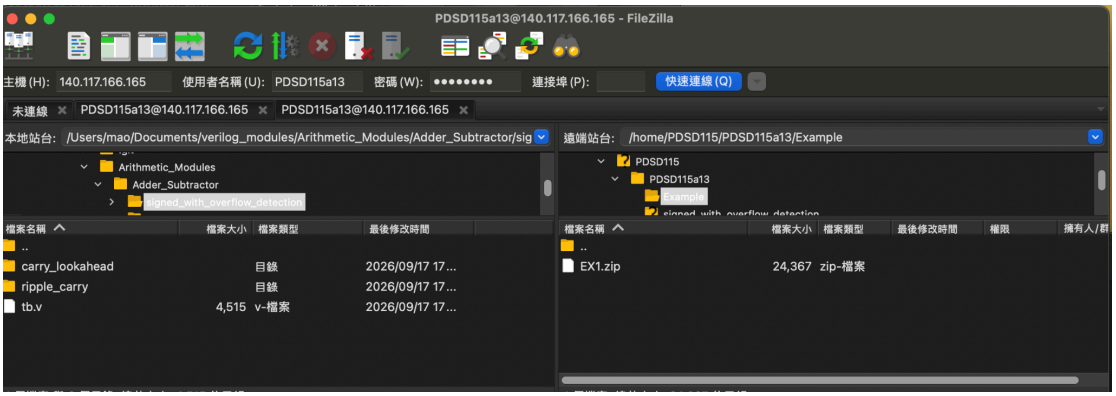
(Also, please include a screenshot.)

5. According to page 23 of the "03-I SOC Server Overview" handout, after entering the "xclock &" command in the terminal, can you see a window displaying the clock icon? ____



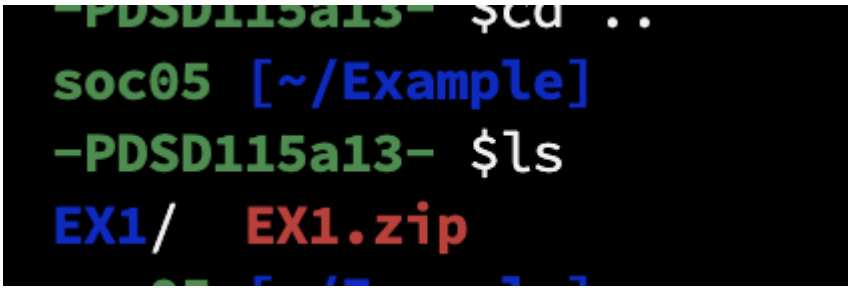
(Also, please include a screenshot.)

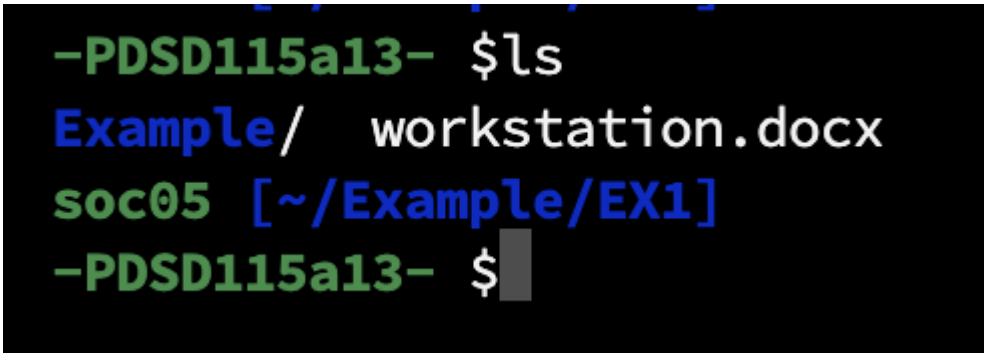
6. According to pages 24 to 26 of the “03-I SOC Server Overview” handout, after completing the FileZilla setup, can you see the folders and files of your workstation account on the remote site? ____
(Also, please include a screenshot.)

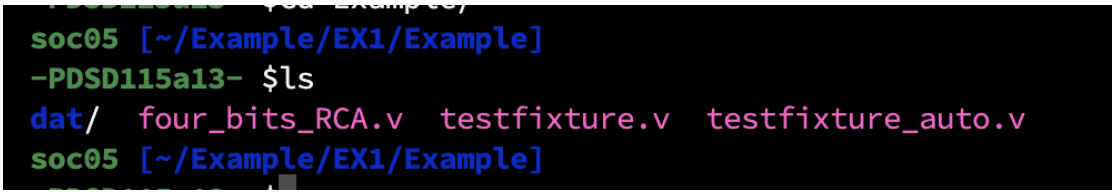


【Phase 2】: Simulation and Verification

1. Download the EX1.zip file from the cyber university, extract it, and upload it to your workstation account using FileZilla. Can you see the EX1 folder on the remote site? ____







(Also, please include a screenshot.)

2. According to page 9 of the “03-II_Operation in Workstation” handout, after entering the command ncverilog four_bits_RCA.v -c in the terminal, does it check if the syntax is correct? ____

```
-PDSD115a13- $ncverilog four_bits_RCA.v -c
ncverilog(64): 15.20-s039: (c) Copyright 1995-2017 Cadence Design Systems, Inc.
file: four_bits_RCA.v
        module worklib.four_bits_RCA:v
                errors: 0, warnings: 0
                Caching library 'worklib' ....... Done
        Elaborating the design hierarchy:
        Building instance overlay tables: ..................... Done
        Generating native compiled code:
                worklib.four_bits_RCA:v <0x389d7681>
                        streams:   0, words:     0
        Building instance specific data structures.
        Loading native compiled code:     ..................... Done
        Design hierarchy summary:
                                Instances  Unique
                Modules:                1       1
                Scalar wires:           1       -
                Vectored wires:         2       -
                Cont. assignments:      0       2
        Writing initial simulation snapshot: worklib.four_bits_RCA:v
soc05 [~/Example/EX1/Example]
-PDSD115a13- $
```

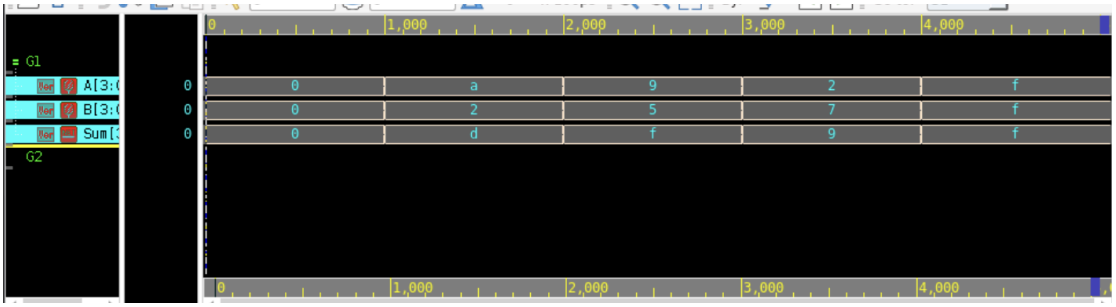
(Also, please include a screenshot.)

3. According to page 11 of the “03-II_Operation in Workstation” handout, after entering the command ncverilog testfixture.v +access+r in the terminal, can you see the simulation results? ____

```
ile: testfixture.v
        module worklib.four_bits_RCA:v
                errors: 0, warnings: 0
        module worklib.textfixture:v
                errors: 0, warnings: 0
                Caching library 'worklib' ....... Done
        Elaborating the design hierarchy:
        Building instance overlay tables: ..................... Done
        Generating native compiled code:
                worklib.four_bits_RCA:v <0x03cf5b86>
                        streams:   2, words:   430
                worklib.textfixture:v <0x3f26330f>
                        streams:   7, words:  8265
        Building instance specific data structures.
        Loading native compiled code:     ..................... Done
        Design hierarchy summary:
                                Instances  Unique
                Modules:                2       2
                Registers:              3       3
                Scalar wires:           2       -
                Vectored wires:         4       -
                Initial blocks:         3       3
                Cont. assignments:      2       2
                Pseudo assignments:     3       3
                Simulation timescale:  100ps
        Writing initial simulation snapshot: worklib.textfixture:v
oading snapshot worklib.textfixture:v ..................... Done
Verdi* Loading libsscore_ius152.so
csim> source /usr/cad/cadence/INCISIV/INCISIVE_15.20.039/tools/inca/files/ncsimrc
csim> run
SDB Dumper for IUS, Release Verdi_X-2025.06, Linux x86_64/64bit, 05/31/2025
C) 1996 - 2025 by Synopsys, Inc.
Verdi* FSDB WARNING: The FSDB file already exists. Overwriting the FSDB file may crash the programs that are using this file.
Verdi* : Create FSDB file '4_bits_adder.fsdb'
Verdi* : Begin traversing the scopes, layer (0).
Verdi* : End of traversing.
                0 Sum= 0 A= 0 B= 0 C_in=0 C_out=0
               10 Sum=13 A=10 B= 2 C_in=1 C_out=0
               20 Sum=15 A= 9 B= 5 C_in=1 C_out=0
               30 Sum= 9 A= 2 B= 7 C_in=0 C_out=0
               40 Sum=15 A=15 B=15 C_in=1 C_out=1
imulation complete via $finish(1) at time 500 NS + 0
/testfixture.v:28 #10 $finish;
csim> exit
oc05 [~/Example/EX1/Example]
PDSD115a13- $
```

(Also, please include a screenshot.)

4. According to pages 14 to 18 of the “03-II_Operation in Workstation” handout, after completing the relevant operations, can you see the waveform simulation results? ____



(Also, please include a screenshot.)

5. Modify Pattern.dat and Expected.dat based on the input values to be tested (C_in, A, B) and the expected output results (C_out, Sum). After entering the command ncverilog testfixture_auto.v +access+r in the terminal, can you see the Totoro icon? ____

```
============================ Simulation START !! ============================


                    Pattern01:  ** Correct!! **
                    Pattern02:  ** Correct!! **
                    Pattern03:  ** Correct!! **
                    Pattern04:  ** Correct!! **
                    Pattern05:  ** Correct!! **


============================ Simulation Finish ! ============================


                 ****************************
                 **                        **        /|__/|
                 **    Congratulations !!  **      / O,O  |
                 **                        **    /_____   |
                 **  Simulation Complete!! **   /^ ^ ^ \  |
                 **                        **  |^ ^ ^ ^ |w|
                 ****************************   \m___m__|_|
```

(Also, please include a screenshot.)

【File Description】: EX1.zip

Folder name	File name		說明
Example	dat	Expected.dat	Output comparison data, which will be used in the automated Testbench
		Pattern.dat	Input test data, which will be used in the automated Testbench
	four_bits_RCA.v		4-bit Adder Circuit
	testfixture.v		Manual Testbench
	testfixture_auto.v		Automated Testbench

【Notes】:

Each person must submit a PDF file of this learning task to the EX1 folder on the FTP, with the submission deadline being before 23:59 on October 4, 2026 (Sunday).